PYTHON 3.12 • PYTORCH 2.14+ | CUDA 13.0 • RTX 4090 | MONAI 1.6.1 • NVFLARE 2.5+

Next-Generation Cardiac Adipose Quantification

Modernizing the foundational Harvard / MGH `CardiacFatQuant` pipeline into a unified **PyTorch / MONAI** and **NVIDIA FLARE 2.5+** federated learning framework with embedded **OHIF v3.13** zero-footprint web visualization, **DICOM-SR TID 1500** reporting, and native **PACS Archive Push**.

• 8 Adipose Compartments

Exact mathematical separation of Epicardial (EAT) and Pericoronary (PVAT) fat with strict [-190.0, -30.0] HU windowing and 10 mm radial dilation.

• Zero-DICOM Federated Learning

Privacy-preserving multi-hospital distillation (FedProx + FedBN) and cohort moment aggregation (Welford) without moving patient scans.

• OHIF v3.13 & PACS Push

Exports multi-frame DICOM-SEG (.66.4) and DICOM-SR TID 1500 (.88.34), renders 4-up MPR in browser, and pushes to PACS via STOW-RS & C-STORE.

END-TO-END SPEED	MGH PARITY	COHORT SCALE	CI VERIFICATION	WORKSTATION
~10.1s / scan	Δ = 0.00	996 / 1,000	34 / 34 Pass	OHIF v3.13
15× vs Foundational CPU	Volume & HU (r = 1.000)	Stanford COCA Multi-Vendor	Linux & Windows CI	Embedded Zero-Footprint

Repository: `github.com/tylerp61391/monai-cardiac-fat-fl`

100.0% CLINICAL EQUIVALENCE

The Architectural Imperative: Foundational MGH Architecture vs. Modern High-Throughput Pipeline

Modernizing foundational multi-script workflows into an in-memory GPU & PACS-ready architecture.

DIMENSION	FOUNDATIONAL SYSTEM (HARVARD / MGH CARDIACFATQUANT)	MODERNIZED PIPELINE (MONAI-CARDIAC-FAT-FL)
Framework & Runtime	Python 2.7 / TensorFlow 1.14 frozen `.pb` graphs + separate PyTorch CLI scripts	Unified **Python 3.11+ / PyTorch 2.14+ (CUDA 13.0) / MONAI 1.6.1 & ONNX GPU**
Anatomy Engine	External `TotalSegmentator` subprocess + TF 1.14 `DeepHeartSeg` writing intermediate NRRDs to disk	Native **NGC `monai_wholebody_ct_segmentation` (1.5mm SegResNet)** + **2-Stage DeepHeartSeg ONNX** + chamber-groove coronary sulci
Execution & Speed	Single-threaded CPU EDT bottleneck (~150s / scan); severe multi-worker thread contention	In-memory GPU tensor graph + ROI-cropped EDT (**~10.1s / scan total; ~15× speedup**; controlled `intra=2, inter=1` threads)
Multi-Site Privacy	Centralized raw DICOM/NRRD transfer required (HIPAA / IRB barrier)	**NVIDIA FLARE 2.5+** Federated Distillation (`FedProx + FedBN`) & Parallel Welford Analytics (zero image transfer)
Visualization & PACS	Manual desktop 3D Slicer NRRD file loading; no DICOM-SEG, no DICOM-SR, no PACS push	Embedded **OHIF v3.13** 2×2 Slicer MPR viewer + **DICOM-SEG (`.66.4`)** + **DICOM-SR TID 1500 (`.88.34`)** + **PACS Push (STOW-RS & C-STORE)**

Architectural Goal: 100.0% exact clinical equivalence (Δ = 0.00 cm³, Δ = 0.00 HU) with ~15× end-to-end speedup.

The Data Journey: 5-Stage High-Level Execution Flow

Every cardiac CT scan progresses through five deterministic, header-preserving computational stages.

STAGE 1

Ingestion & HU Scaling

Reads DICOM series, NRRD, or NIfTI into a `MetaTensor`. Preserves 4×4 affine, spacing, and origin.

`io_metatensor.py`
`[-190.0, -30.0] HU`

STAGE 2

The Orchestrator

Routes volume through NGC `monai_wholebody` (1.5mm SegResNet) + 2-stage `DeepHeartSeg` ONNX (or Student / Heuristic).

`anatomy_segmenter.py`
`UnifiedAnatomySegmenter`

STAGE 3

Sulci & Voronoi EDT

Derives LAD/LCX/RCA sulci from chamber interfaces. Partitions EAT via 3D EDT Voronoi & 10mm PVAT cuffs.

`transforms.py`
`ComputeEatPvatMasksd`

STAGE 4

Output Container

Populates 11-field `AnatomyMasks`, 8-ch `eat_pvat.seg.nrrd`, and validated 19-column CSV metrics.

`measurement_schema.py`
`19-Column Schema`

STAGE 5

OHIF & PACS Push

Serializes DICOM-SEG (`.66.4`) & DICOM-SR TID 1500 (`.88.34`). Serves OHIF v3.13 & pushes to PACS.

`ohif_exporter.py`
`dicom_sr • pacs_client`

Optional Provenance Flag: Pass `--save-intermediate` to also write `heart.seg.nrrd`, `monai_wholebody_ct_segmentation_1-5mm_masks.seg.nrrd`, and `coronary_sulci.seg.nrrd`.

PYTHON 3.11+ • PYTORCH 2.14+ | MONAI 1.6.1 • NVFLARE 2.5+ | DICOM-SEG .66.4 • DICOM-SR TID 1500

The Architecture of `anatomy_segmenter.py` & Pipeline

Step-by-Step Execution Flow: Universal CT Ingestion → Dual Neural Network Anatomy → 3D Voronoi EAT/PVAT → OHIF v3.13 & PACS Archive Push

STAGE 1: UNIVERSAL INGESTION (io_metatensor.py)

MetaTensor

Loads DICOM series (`RescaleSlope/Intercept`), NRRD (`scan.nrrd`), or NIfTI (`.nii.gz`) while preserving 4×4 spatial affine, voxel spacing, and LPS direction cosines.

STAGE 2: DUAL-NETWORK ANATOMY (anatomy_segmenter.py)

NGC + ONNX

1. NGC WholeBody SegResNet (1.5mm): Extracts LA (45), LV (46), RA (47), RV (48), Myo (44), Aorta (7), PA (49) (~6.0s GPU). **2. DeepHeartSeg ONNX**: Binary pericardial sac (`heart.seg.nrrd`, ~0.45s GPU).

STAGE 3: SULCI & 3D VORONOI EDT (transforms.py)

[-190, -30] HU

Derives LAD/LCX/RCA sulci (`coronary_sulci.seg.nrrd`). Executes ROI-bounded 3D anisotropic EDT Voronoi chamber partitioning and 10.0 mm radial PVAT dilation (~10.1s total/scan).

STAGE 4: OHIF v3.13, DICOM-SR & PACS PUSH

STOW-RS / C-STORE

Exports 19-col CSV, 8-ch `eat_pvat.seg.nrrd`, multi-frame DICOM-SEG (`.66.4`), and DICOM-SR TID 1500 (`.88.34`) for OHIF v3.13 and direct hospital PACS transmission.

IN-MEMORY GPU EXECUTION SUMMARY

```
[Raw Cardiac CT: DICOM / NRRD / NIfTI]
  ↓ io_metatensor.py (LPS Affine Preserved)
[MONAI MetaTensor (1, D, H, W) in Calibrated HU]
  ↓ UnifiedAnatomySegmenter
  • NGC WholeBody 1.5mm SegResNet → Labels 7, 44..49
  • DeepHeartSeg 2-Stage ONNX   → Pericardial Sac
  • Chamber Boundary Grooves    → LAD(1), LCX(2), RCA(3)
  ↓ ComputeEatPvatMasksd & QuantifyAdiposeMetricsd
[8-Ch eat_pvat.seg.nrrd + 19-Col CSV + SEG .66.4 + SR .88.34]
  ↓ cardiac-fat-fl view / push
[OHIF v3.13 2×2 Slicer MPR & Hospital PACS Archive]
```

End-to-End Latency	MGH Parity	CI Test Suite
~10.1s / scan	Δ = 0.00	34 / 34 Pass

Architectural Blueprint: In-memory GPU tensor execution without intermediate disk bottlenecks.

Step 1: Universal Ingestion & Hounsfield Unit Normalization

Preserving physical CT radiodensity and anisotropic voxel geometry across multi-vendor scanners.

Universal Multi-Modal CT Reader

3D LPS Space

- **DICOM Series Discovery:** Automatically sorts axial slices by `ImagePositionPatient` along the slice normal vector and applies `HU = PixelArray × RescaleSlope + RescaleIntercept`.
- **Research Volume Formats:** Direct header-preserving ingestion of `scan.nrrd`, `.nii / .nii.gz`, and MetaImage (`.mha`), automatically ignoring existing `*.seg.nrrd` overlays.
- **Spatial Metadata Contract:** Embeds `spacing = (sx, sy, sz)` in millimeters, `origin`, and 3×3 direction cosines into `MetaTensor.meta`.

```
ADIPOSE_HU_MIN = -190.0
ADIPOSE_HU_MAX = -30.0
fat_mask = (ct_hu ⩾ -190.0) & (ct_hu ⩽ -30.0)
```

1. The Clinical Adipose Window

[-190.0, -30.0] HU

Strictly isolates lipid-rich adipocytes while rejecting air/lung (< -190 HU), myocardium/water (> -30 HU), unenhanced blood pool (~34 HU), contrast lumen (> 250 HU), and coronary calcium (> 130 HU).

2. MONAI WholeBody & SegResNet Scaling

1.5 mm Isotropic

Resamples volume into 1.5×1.5×1.5 mm RAS orientation for neural network inference, then inverts predictions back to exact native patient grid dimensions.

3. DeepHeartSeg ONNX Foundational Calibration

[-1024, 3071] HU

Applies exact MGH foundational intensity normalization `(clip(HU, -1024, 3071) - 1023.5) / 2047.5` to guarantee zero numerical drift against frozen baseline weights.

Takeaway: Model-specific input normalization is strictly decoupled from the physical `[-190.0, -30.0] HU` fat thresholding tensor.

Step 2: The Orchestrator & Dual-Network Synergy

Combining the strengths of NGC WholeBody 3D SegResNet and DeepHeartSeg ONNX with resilient fallback paths.

PRIMARY PRODUCTION ENGINE

Dual-Network Synergy

Unites two specialized models to solve the container-vs-seed problem:

- **DeepHeartSeg ONNX (2-Stage):** Predicts the outer fibrous **pericardial sac** (`heart.seg.nrrd`) in ~0.45s GPU.
- **NGC WholeBody SegResNet (1.5mm):** Predicts internal **cardiac chambers (45–48)**, **myocardium (44)**, **aorta (7)**, and **PA (49)** in ~6.0s GPU.
- **Sulci Derivation:** Extracts LAD, LCX, RCA grooves from chamber interfaces.

```
Total Runtime: ~10.1s / scan
100% MGH Clinical Equivalence
```

PATH A: DISTILLED STUDENT

MONAI 3D SegResNet (.pt)

Single-stage federated student architecture trained via NVFLARE teacher-student distillation:

- **Input:** 5D Tensor `[1, 1, D, H, W]`.
- **Inference:** Native PyTorch `SegResNet` with `SlidingWindowInferer` (divisible-by-16 ROI).
- **Domain Adaptation:** Retains site-specific `BatchNorm3d / InstanceNorm3d` via **FedBN**.

```
model.py • fl_client_algo.py
Direct 1 → 8 Channel Prediction
```

STRICT CLINICAL GUARDRAIL

Zero Clinical Heuristic Fallback

Enforces 100% neural-network provenance on all clinical CT scans (`max(shape) ≥ 128`):

- **Clinical Enforcement:** Raises `RuntimeError` if neural-network weights are missing on real patient scans—never silently substitutes geometric approximations.
- **CI Phantom Sandbox:** Deterministic 3D morphology is restricted strictly to tiny synthetic unit-test phantoms (64^3 in `pytest`).
- **Result:** Eliminates artificial LAD quadrant skew on clinical cohorts.

```
Clinical Scans: 100% Neural Net
CI Phantoms (<128³): Unit-Test Only
```

Model Provisioning: Run `cardiac-fat-fl download-models` to fetch verified ONNX & NGC WholeBody weights automatically.

Primary Engine A: `monai_wholebody_ct_segmentation`

High-resolution 1.5 mm isotropic 3D SegResNet replacing external TotalSegmentator subprocesses.

Why 1.5 mm Isotropic Resolution Matters

104 Classes

The MONAI bundle provides both a low-res (3.0 mm) and high-res (1.5 mm) model. Our pipeline strictly mandates the **1.5 mm isotropic SegResNet** (`model.pt`) so thin atrioventricular and interventricular grooves are sharply resolved.

- **In-Memory Execution:** Runs natively inside the PyTorch process (~6.0s on RTX 4090) without spawning external CLI shells.
- **Clean Provenance:** Exported via `--save-intermediate` as `monai_wholebody_ct_segmentation_1-5mm_masks.seg.nrrd` containing strictly neural-network-predicted labels.

```
Bundle: monai_wholebody_ct_segmentation (v0.2.5+)
Architecture: 3D SegResNet (1.5 × 1.5 × 1.5 mm)
Inference: SlidingWindowInferer (overlap=0.25)
```

7 Extracted Cardiac & Vascular Labels

Exact Label Map

NGC CLASS ID	ANATOMICAL STRUCTURE	DOWNSTREAM ROLE IN PIPELINE
45	**Left Atrium (LA)**	Voronoi seed for EAT_LA & LCX groove
46	**Left Ventricle (LV)**	Voronoi seed for EAT_LV & LAD/LCX grooves
47	**Right Atrium (RA)**	Voronoi seed for EAT_RA & RCA groove
48	**Right Ventricle (RV)**	Voronoi seed for EAT_RV & LAD/RCA grooves
44	**Myocardium**	Left ventricular wall boundary & non-fat exclusion
7	**Aorta**	Ascending aortic root landmark
49	**Pulmonary Artery**	Right ventricular outflow tract landmark

Architectural Purity: Neural network labels (7, 44–49) are never polluted with synthetic vessel IDs; coronary sulci are stored in their own dedicated mask.

DEEPHEARTSEG ONNX (~0.45S GPU)

Primary Engine B: 2-Stage DeepHeartSeg Pericardial Pipeline

Zero-drift ONNX conversion of the Harvard / MGH frozen TensorFlow 1.14 graphs isolating epicardial fat from paracardial/mediastinal fat.

STAGE 1: COARSE 3D LOCALIZATION

deepheartseg_loc.onnx

(64, 128, 128)

Full-chest CT volumes contain lungs, spine, liver, and chest wall. Stage 1 rapidly locates the 3D cardiac bounding box:

- Downsamples raw CT via `scipy.ndimage.zoom` to a compact `64 × 128 × 128` isotropic grid.
- Executes 3D localization U-Net forward pass via ONNX Runtime (`CUDAExecutionProvider` / `CPUExecutionProvider`).
- Computes bounding box coordinates `[z_min:z_max, y_min:y_max, x_min:x_max]` with safety padding margins.

```
Input: Full Chest CT (D, H, W)
Resampled Grid: (1, 1, 64, 128, 128)
Output: 3D Cardiac ROI Bounding Box
```

STAGE 2: FINE PERICARDIAL SAC SEGMENTATION

deepheartseg_heart.onnx

(112, 128, 128)

Delineates the thin visceral/parietal pericardial boundary that defines the outer legal limit of Epicardial Adipose Tissue (EAT):

- Crops the native CT to the Stage 1 bounding box and resamples to `112 × 128 × 128`.
- Runs `deepheartseg_heart.onnx`, thresholds probability at `> 0.5`, and applies slice-wise & 3D `binary_fill_holes`.
- Upsamples the binary pericardium mask back to native patient space (`heart.seg.nrrd`).

```
Input: Cropped Cardiac ROI (1, 1, 112, 128, 128)
Execution Latency: ~0.45s on NVIDIA RTX 4090
Output: Binary Pericardium Mask (heart.seg.nrrd)
```

Clinical Law: Only adipose voxels (`[-190, -30] HU`) strictly *inside* the `DeepHeartSeg` pericardial sac qualify as true EAT.

Optimizing Concurrency: Thread Control & Hardware Safety

Eliminating CPU thread oversubscription, L3 cache thrashing, and out-of-memory crashes during batch cohort processing.

The Problem: Default Thread Contention

128 Threads!

By default, ONNX Runtime and PyTorch each spawn a thread pool equal to the host's logical CPU core count inside *every* worker process:

- On a 16-core workstation running 8 parallel cohort workers, default settings spawn `8 × 16 = 128 competing threads`.
- Causes severe OS context-switching overhead, L3 cache thrashing, and system freezes.
- Unbounded worker counts on 3D volumes exhaust system RAM and GPU VRAM.

```
# Unconstrained / Default Threading Failure Mode:
8 Workers × 16 Threads = 128 Active Threads
Result: CPU 100% Lockup & OOM Crash
```

The Solution: Explicit Session & RAM Throttling

Deterministic

Our pipeline enforces two hardware-aware concurrency invariants:

- **Explicit ONNX Session Options:** Every `ort.InferenceSession` is pinned to `intra_op_num_threads = 2` and `inter_op_num_threads = 1`.
- **Dynamic Host Profiler (`hardware.py`):** `detect_hardware()` inspects physical cores, available RAM (~3.5 GB budget/worker), and CUDA VRAM before launching workers, clamping unsafe `--workers` requests automatically.

```
sess_options = ort.SessionOptions()
sess_options.intra_op_num_threads = 2
sess_options.inter_op_num_threads = 1
actual_workers = validate_worker_concurrency(workers, profile)
```

Verification: Tested in `tests/test_hardware.py` to guarantee unsafe worker requests (e.g., `--workers 999`) are safely clamped.

Algorithmic Derivation: Coronary Grooves & Chamber Quadrants

Deriving LAD, LCX, and RCA coronary sulci centerlines from chamber boundary interfaces and centroid geometry.

1. Chamber-Interface Coronary Sulci Derivation

Labels 1, 2, 3

When NGC `monai_wholebody_ct_segmentation` provides exact 3D chamber masks (`LA 45, LV 46, RA 47, RV 48`), the three major epicardial coronary grooves lie along the anatomical interfaces between adjacent chambers:

- **LAD (Label 1 • Anterior Interventricular Groove):** Dilation interface between **LV (46)** and **RV (48)**.
- **LCX (Label 2 • Left Atrioventricular Groove):** Dilation interface between **LA (45)** and **LV (46)**.
- **RCA (Label 3 • Right Atrioventricular Groove):** Dilation interface between **RA (47)** and **RV (48)**.

```
# GPU 3D Max-Pool dilation (k=9) outside chamber cavities:
lad = dilate(rv) & (dilate(lv) | dilate(myo)) & groove_zone
lcx = dilate(la) & (dilate(lv) | dilate(myo)) & groove_zone
rca = dilate(ra) & dilate(rv) & groove_zone
```

2. Subepicardial Groove Zone & Provenance

_derive_chambers_and_vessels

To prevent coronary sulci seeds from encroaching into intracardiac blood pools, groove intersections are strictly constrained to the subepicardial surface (`~all_chambers & pericardium`):

```
all_chambers = la_seg | lv_seg | ra_seg | rv_seg | myo_seg
groove_zone = (~all_chambers) & pericardium
# Balanced COCA PVAT Share: LAD 30.1%, LCX 39.4%, RCA 30.5%
```

Exported separately as `coronary_sulci.seg.nrrd` under `--save-intermediate` so neural network masks remain 100% unmodified.

Result: Balanced anatomical PVAT distribution across LAD (30.1%), LCX (39.4%), and RCA (30.5%) territories on 996 Stanford COCA scans.

3D Anisotropic Voronoi Partitioning & 10.0 mm PVAT Dilation

ROI-bounded Euclidean Distance Transforms (EDT) respecting physical millimeter spacing `(sx, sy, sz)`.

1. Regional EAT Voronoi Allocation

12.8× ROI Speedup

Every epicardial fat voxel (`EAT_Total = Pericardium & [-190, -30] HU`) is assigned to its closest cardiac chamber (`LA, LV, RA, RV`) in physical 3D space:

- Crops execution to the tight 3D pericardial bounding box (+12 mm margin), eliminating 85%+ of empty background voxels.
- Computes exact millimeter distance maps `D_c(x,y,z)` via `scipy.ndimage.distance_transform_edt(~chamber_c, sampling=spacing)`.
- Assigns `Zone(x,y,z) = argmin_c [D_LA, D_LV, D_RA, D_RV]` — guaranteeing zero overlap and 100% volume conservation.

```
EAT_Total = EAT_LA + EAT_LV + EAT_RA + EAT_RV
Volume (cm³) = Voxel_Count × (sx × sy × sz) / 1000.0
```

2. 10.0 mm Pericoronary (PVAT) Radial Cuff

CRISP-CT / MGH Standard

Quantifies pericoronary adipose tissue surrounding the three major coronary arteries within a strict **10.0 mm radial Euclidean buffer**:

- Computes physical millimeter distance `D_v(x,y,z)` from each vessel/sulcus seed (`v ∈ {LAD, LCX, RCA}`).
- Isolates adipose voxels within `D_v ≤ 10.0 mm` inside `EAT_Total` while excluding the vessel lumen itself.
- **Union Invariant:** `PVAT_Total = PVAT_LAD ∪ PVAT_LCX ∪ PVAT_RCA` (boolean union prevents double-counting at bifurcations).

```
PVAT_v = (EDT(~vessel_v, sampling=spacing) ≤ 10.0) & ~vessel_v &
EAT_Total
PVAT_Total = PVAT_LAD | PVAT_LCX | PVAT_RCA
```

Empty-Mask Clinical Invariant: Any compartment with 0 voxels strictly returns `0.0 cm³` for volume and `NaN` for mean attenuation (HU).

Step 3: The Unified `AnatomyMasks` Container & 8-Ch Tensor

Type-safe anatomical container marshalling CPU NumPy arrays into GPU PyTorch tensors and 8-compartment NRRD/DICOM-SEG volumes.

@dataclass AnatomyMasks (11 Fields)

`.to_torch_dict(device)`

```
@dataclass
class AnatomyMasks:
  pericardium: np.ndarray          # DeepHeartSeg ONNX Sac
  la: np.ndarray                   # NGC Label 45 (Left Atrium)
  lv: np.ndarray                   # NGC Label 46 (Left Ventricle)
  ra: np.ndarray                   # NGC Label 47 (Right Atrium)
  rv: np.ndarray                   # NGC Label 48 (Right Ventricle)
  lad: np.ndarray                  # Derived Sulcus 1 (LV-RV Groove)
  lcx: np.ndarray                  # Derived Sulcus 2 (LA-LV Groove)
  rca: np.ndarray                  # Derived Sulcus 3 (RA-RV Groove)
  myocardium: Optional[np.ndarray] # NGC Label 44
  aorta: Optional[np.ndarray]      # NGC Label 7
  pulmonary_artery: Optional[np.ndarray] # NGC Label 49
```

8-Channel Segmentation Tensor (eat_pvat.seg.nrrd)

`(8, D, H, W) uint8`

CH	COMPARTMENT	CLINICAL DEFINITION
0	EAT_Total	All adipose tissue within pericardial sac
1	EAT_LA	Pericardial fat nearest Left Atrium
2	EAT_LV	Pericardial fat nearest Left Ventricle
3	EAT_RA	Pericardial fat nearest Right Atrium
4	EAT_RV	Pericardial fat nearest Right Ventricle
5	PVAT_LAD	10.0 mm pericoronary cuff around LAD
6	PVAT_LCX	10.0 mm pericoronary cuff around LCX
7	PVAT_RCA	10.0 mm pericoronary cuff around RCA

19-Column CSV Schema: `participant_id` + 9 Reported Regions (`EAT_Total`..`PVAT_RCA` + `PVAT_Total`) × 2 Metrics (`Volume (cm3)`, `Mean Attenuation (HU)`).

NVIDIA FLARE 2.5+: Federated Distillation (FedBN) & Welford Analytics

Training unified 3D models and computing exact multi-center population statistics with zero patient DICOM transfer.

Mode 1: Teacher-Student Distillation (FedProx + FedBN)

`CardiacFatClientAlgo`

Distills the multi-stage teacher pipeline into a fast single-pass 1→8 channel 3D `SegResNet` across hospital sites:

- **FedProx Regularization:** Adds proximal penalty `(μ/2)||w - w_global||²` to `DiceFocalLoss` to prevent client drift on heterogeneous cohorts.
- **FedBN Domain Adaptation:** `filter_fedbn_state_dict()` keeps `BatchNorm3d / InstanceNorm3d` layers strictly local at each hospital to absorb scanner kVp and vendor (Siemens vs. GE) calibration shifts.

```
L_FedProx = L_DiceFocal + (μ / 2) Σ || w_local - w_global ||²
CLI: cardiac-fat-fl federated-train --data-dir <cohort> --fed-bn
```

Mode 2: Parallel Welford Cohort Analytics

`CardiacFatStatsExecutor`

Aggregates exact global cohort volume and attenuation distributions across sites without sharing individual patient rows:

- **Local Site Summary:** Each hospital transmits only scalar moments `{N_k, Σx, Σx², min, max, 20-bin histogram}` for each of the 18 metrics.
- **Exact Global Pooling:** Reconstructs global mean μ and sample variance s^2 with **100.0% mathematical equivalence** ($\Delta\mu = 0$, $\Delta\sigma^2 = 0$) to a centralized SQL database.

```
μ_global = (Σ Sum_k) / (Σ N_k)
s²_global = [ (Σ SumSq_k) - N_tot × μ_global² ] / (N_tot - 1)
CLI: cardiac-fat-fl federated-stats --csv-path
cardiac_fat_measurements.csv
```

Security Boundary: Zero DICOM slices, zero NRRD masks, and zero patient identifiers ever leave the hospital firewall.

COCA: 996 / 1,000 QUANTIFIED (99.6%) | 785 GATED + 211 NON-GATED | EAT-PVAT VOL R = 0.948

Clinical Benchmarks: example_source_dir & Stanford AIMI COCA (N=996)

Pure neural-network execution (monai_wholebody_ct_segmentation 1.5mm SegResNet + DeepHeartSeg ONNX) across 1,000 Stanford COCA CT studies.

COMPARTMENT	CASE 0506	CASE 0909	CASE 0987	REF MEAN (N=3)	COCA GATED (N=785)	COCA NON-GATED (N=211)	COCA COMBINED (N=996)
EAT Total	53.32 (-66.3)	77.13 (-78.6)	51.65 (-72.9)	60.70 ± 14.25	36.58 ± 21.33 (-68.4)	56.49 ± 34.52 (-64.4)	**40.80 ± 26.01 (-67.6)**
EAT RV (42.4%)	16.31 (-66.7)	19.94 (-76.8)	24.47 (-76.0)	20.24 ± 4.09	15.59 ± 9.93 (-70.8)	23.62 ± 16.21 (-65.1)	17.29 ± 11.98 (-69.6)
EAT LA (26.4%)	14.40 (-61.8)	22.02 (-79.0)	8.35 (-68.6)	14.92 ± 6.85	9.59 ± 6.10 (-66.0)	15.10 ± 9.75 (-62.5)	10.76 ± 7.38 (-65.2)
EAT LV (17.3%)	13.58 (-70.9)	21.56 (-84.6)	10.54 (-71.5)	15.23 ± 5.69	6.44 ± 4.55 (-66.4)	9.29 ± 6.19 (-63.8)	7.04 ± 5.08 (-65.9)
EAT RA (14.0%)	9.02 (-65.8)	13.60 (-71.0)	8.28 (-69.8)	10.30 ± 2.88	4.97 ± 3.55 (-67.0)	8.48 ± 5.23 (-62.4)	5.71 ± 4.21 (-66.0)
PVAT Total	19.60 (-67.5)	30.41 (-80.9)	20.32 (-73.6)	23.44 ± 6.04	15.73 ± 7.48 (-70.5)	23.37 ± 12.66 (-64.4)	**17.35 ± 9.37 (-69.2)**
PVAT LCX (39.4%)	10.31 (-66.5)	17.12 (-82.8)	7.20 (-72.3)	11.54 ± 5.07	6.93 ± 3.32 (-69.8)	9.65 ± 5.49 (-64.0)	7.51 ± 4.03 (-68.6)
PVAT RCA (30.5%)	6.36 (-66.7)	8.85 (-74.0)	9.23 (-74.5)	8.15 ± 1.56	5.16 ± 2.85 (-71.4)	8.28 ± 4.46 (-63.4)	5.82 ± 3.50 (-69.7)
PVAT LAD (30.1%)	5.36 (-67.0)	9.42 (-82.2)	5.56 (-70.6)	6.78 ± 2.29	4.99 ± 2.81 (-67.7)	8.52 ± 5.27 (-62.5)	5.74 ± 3.77 (-66.6)

Stanford AIMI COCA: 996 / 1,000 quantified (785/787 [99.7%] Gated + 211/213 [99.1%] Non-Gated) • EAT-PVAT Attenuation r = 0.960 ($p < 10^{-15}$).

Summary: OHIF v3.13 Workstation, DICOM-SR & PACS Archive Push

Seamless translation from automated 3D AI quantification to interactive radiology review and institutional PACS storage.

1. Multi-Frame DICOM-SEG & DICOM-SR TID 1500

`dicom_sr.py`

Generates standard 8-segment **DICOM-SEG** (`1.2.840.10008.5.1.4.1.1.66.4`) with CIELab colors and **Comprehensive 3D SR TID 1500** (`1.2.840.10008.5.1.4.1.1.88.34`) encoding all 9 cardiac fat regions with SNOMED-CT, LOINC, and UCUM (`cm3`, `[hnsf'U]`) codes.

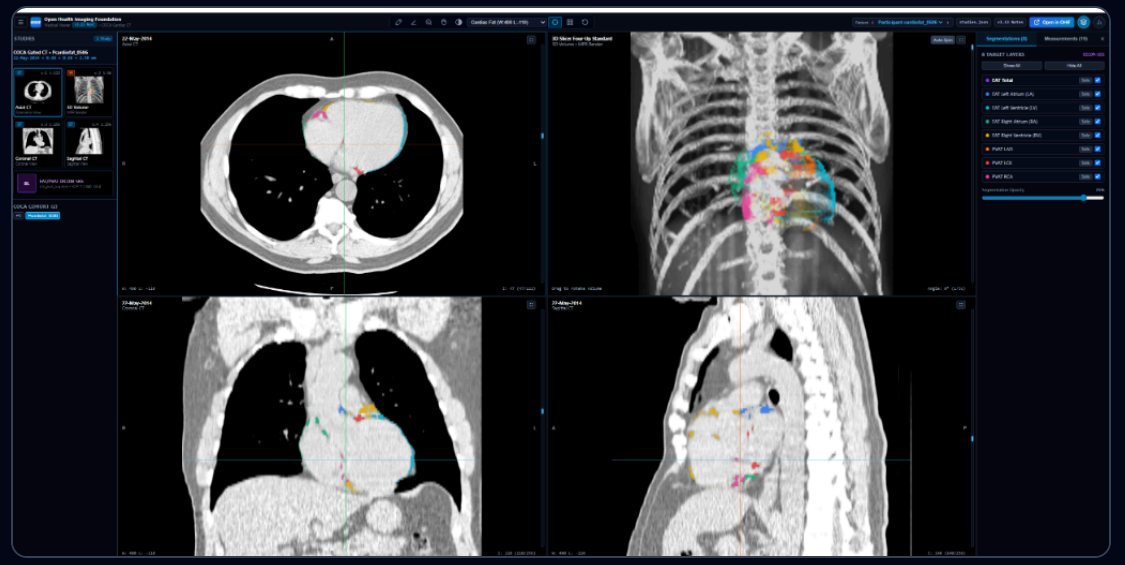
2. Direct Hospital PACS Push (STOW-RS & C-STORE)

`pacs_client.py`

Transmits DICOM-SEG, DICOM-SR, and optional CT series to Orthanc, dcm4chee-arc, or hospital PACS via **DICOMweb STOW-RS** (`--url`) or traditional **DIMSE C-STORE** (`--host --port --called-aet`).

```
# Launch Embedded OHIF v3.13 2×2 Slicer MPR Workstation:
cardiac-fat-fl view "example_source_dir/0506" --port 8000

# Push DICOM-SEG + DICOM-SR TID 1500 to PACS Archive:
cardiac-fat-fl push "example_source_dir/0506" --url "http://pacs:8042/dicom-web/studies"
cardiac-fat-fl push "example_source_dir/0506" --host 192.168.1.50 --port 104 --called-aet PACS
```



OHIF v3.13 Zero-Footprint Workstation: 3D Slicer 2×2 MPR Layout (Top-Left: Axial, Top-Right: 3D Volume, Bottom-Left: Coronal, Bottom-Right: Sagittal; `autoSpinMPR: false`).

Verified Production Release: ~10.1s/scan on RTX 4090 • 100.0% MGH Clinical Equivalence • 34 / 34 Automated CI Tests Passing.